

Synthetic data generation using building information models

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ABSTRACT

Infrastructure scene understanding from image data aids diverse applications in construction and maintenance. Recently, deep learning models have been employed to extract information regarding infrastructure from visual data. The performance of these models depends significantly on the volume of training data. However, preparing the training data is time-consuming and laborious, as it entails labeling numerous images. To address this issue, this paper proposes a method for generating high-quality synthetic data that includes the automatic annotation of infrastructure scenes. The method consists of three steps: 1) translating building information model (BIM) images into real-world images, 2) automatically labeling them using the spatial information contained in the BIM to generate various synthetic datasets, and 3) splicing the selected synthetic datasets together to form the final synthetic dataset. The Mask R-CNN models trained with building and bridge synthetic data achieved average precisions of 71.6% and 84.9%, respectively.

1. Introduction

In the architecture, engineering, construction, and facility management (AEC & FM) industry, the scene understanding of civil infrastructure is crucial for carrying out diverse vision-based automated tasks [1]. Understanding infrastructure scenes involves obtaining the location, shape, color, and quantity information of the physical objects constituting a construction structure. This information can be employed for safety monitoring [2], progress monitoring [3,4], and structural inspection [5–9].

A significant amount of research has been conducted on the understanding of the civil infrastructure scene. In the past decades, methods of extracting features, such as color, texture, and shape, that can distinguish target infrastructure from other objects have been studied using image processing and machine learning techniques. More recently, with the advancement of computers, such as the advent of GPUs with improved speed, deep learning models, particularly those that employ convolutional neural networks (CNNs), have been applied for contextual awareness of the infrastructure scene. Such applications require the collection of a substantial amount of labeled image data. To this end, large-scale publicly available datasets such as the Scene Understanding (SUN) database [10], Common Objects in Context (COCO) dataset [11], Cityscapes dataset [12], and ADE20K [13] have been created. However, data regarding civil infrastructure scenes are lacking. Thus, there is a vital requirement for large-scale annotated image data regarding the

infrastructure scene.

The diversity of the types, shapes, and materials of the objects that constitute infrastructure complicates implementing a recognition model that performs well for all infrastructure scenes. Two challenges are faced while preparing data for infrastructure scene understanding: The first is that image labeling is performed manually, which is time-consuming and costly. Image labeling for object recognition is the task of marking the area containing the object using a polygon and specifying the class of the object. Marking a polygon requires a dozen clicks on a single object. The Google Cloud data labeling service charges \$257 to generate 1000 polygons for instance segmentation [14]. Cordts et al. [12] stated that each fine annotation of a single cityscape image took more than 90 min. The second challenge is that domain knowledge is required to label the physical objects that constitute the infrastructure. Expertise is required to identify the area and class of the objects in an infrastructure scene image. For example, skill is entailed in determining a color code that identifies the contents of a pipe in a plumbing system. The labeler may need additional training to acquire this knowledge.

The challenge of image labeling can be addressed by utilizing building information models (BIMs) with comprehensive information regarding civil infrastructure. BIMs include a variety of information, termed resources, such as information regarding the geometry and material of all components constituting the AEC product. Therefore, pixel-level labeling can be performed for the images obtained by capturing the views of BIMs using a virtual camera in BIM software.

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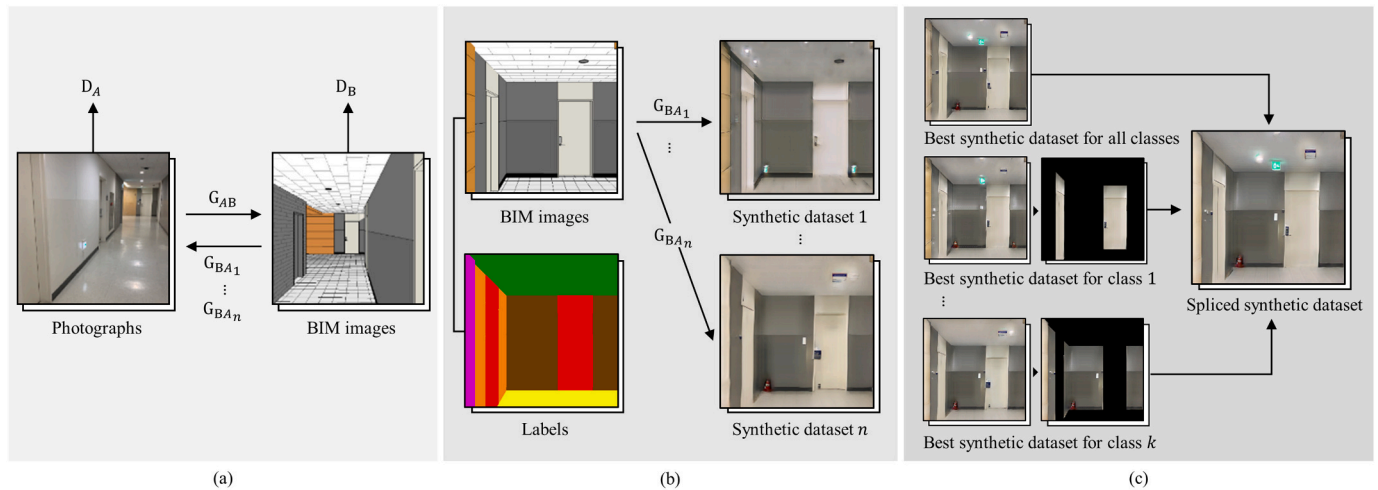


Fig. 1. Research framework: (a) image-to-image translation, (b) synthetic data generation, and (c) synthetic data splice, where n is the number of G_{BA} and k is the number of classes.

However, differences in color and texture between BIM and real-world images—mainly photographs—cause differences in spatial features that serve as the training criteria of CNNs [15]. Therefore, for BIM images to be used as training data for the analysis of photographs, these must be transformed into a photographic style.

The generative adversarial networks (GANs) [16] architecture is presently being studied as a novel data generation method. It is used to create synthetic images, such as handwritten images [16] and human facial images [17], that are similar to the corresponding actual images. The GANs also facilitate image transformation, such as that of grayscale images into color images and day scenes into night scenes [18]. Cycle-consistent adversarial networks (CycleGAN) [19] are models that map images from two different domains. In this paper, we propose a CycleGAN-based method to generate synthetic data for the improved training of deep learning models for the infrastructure scene understanding. The research framework is depicted in Fig. 1. First, as depicted in Fig. 1(a), multiple generators, G_{BA} s, that perform the translation from BIM images to photographs are acquired. Next, G_{BA} s generate synthetic images with labels from the BIM images as shown in Fig. 1(b). Subsequently, experiments are performed using mask regions with CNN features (Mask R-CNN) [20] to assess the efficiency of each synthetic dataset for the training of CNNs for object recognition. Finally, as illustrated in Fig. 1(c), the synthetic datasets that achieve the highest performance for each category of objects are selected as the components of the final dataset. The final synthetic dataset is constructed by splicing the pixels from the selected synthetic datasets. In the experiments, the performance of each Mask R-CNN trained on photographs, BIM images, and synthetic datasets was evaluated. The contribution of this study lies in the creation of a novel method to automatically generate annotated synthetic infrastructure scene images that significantly improve the performance of instance segmentation models.

2. Related work

In the construction industry, supervised learning has emerged as a means for automating existing construction tasks. Therefore, the creation of a large number of data is in the common interest of civil engineers, and efforts are being made to create a substantial amount of publicly available construction object datasets. One of the outcomes of such efforts is Structural ImageNet [21,22]. Gao et al. [21] and Gao and Mosalam [22] collected over 10,000 pre-disaster and post-disaster images from online archives and image search engines to create a dataset of structural damage images referred to as Structural ImageNet. Qurishee et al. [23] created a concrete crack detection dataset consisting of 2088

crack and noncrack images and a pavement crack detection dataset consisting of 2620 images. Wu et al. [24] generated a dataset on construction workers, referred to as GDUT-HWD, that consists of 3174 images for hardhat detection. Kim et al. [25] generated a construction equipment dataset, referred to as the AIM dataset, consisting of 2920 images for equipment classification. Despite these efforts, and considering that ImageNet [26], a representative open dataset in the field of computer vision, contains 21,000 classes and 14 million images, datasets in the civil engineering field are still inadequate in terms of the diversity and quantity of construction objects.

To circumvent the lack of datasets, methods that use a three-dimensional (3D) model or a style transfer method to generate synthetic data have been developed. Soltani et al. [27] proposed a method for generating automatically labeled synthetic data using a 3D excavator model and construction site background images to improve the performance of construction equipment detectors. Kim and Kim [28] reconstructed a 3D model of a concrete mixer truck using the multiview stereo algorithm to generate synthetic data. Further, with the rapid increase in the prevalence and effectiveness of BIM, research is underway to generate synthetic data from BIM. Acharya et al. [29] and Acharya et al. [30] used the indoor images derived from BIM as training data of a deep learning network that estimates the pose of a camera. Acharya et al. [29] used pre-processed images to extract edges from BIM images for fine-tuning of CNN, and Acharya et al. [30] used BIM images rendered in five different ways to train a network consisting of a Bayesian CNN and a long short-term memory layer. Ma et al. [31] used synthetic point clouds obtained from BIM as the training data for point cloud segmentation networks. To create new images by combining the content and style of original images, Gatys et al. [32] proposed an algorithm to change a white noise image to a resultant image with the content and style of the target image using CNN. Johnson et al. [33] used an image transformation network with a deep residual CNN for the style transfer of multiple images.

The GAN [16] has been proposed for generating images that are similar to real images. It is a deep learning model in which the generative model G and discriminative model D , defined by multilayer perceptrons, are simultaneously trained using the adversarial method. G is trained to generate images with features similar to real images, and D is trained to distinguish between real and fake images. Radford et al. [17] proposed a deep convolutional GAN (DCGAN) by applying CNN architectures and a series of constraints to the GAN architecture. The generator and discriminator of the DCGAN were composed of fractional-strided and strided convolutions, respectively. With the advent of DCGAN, the GAN has been applied in various fields. As an application of

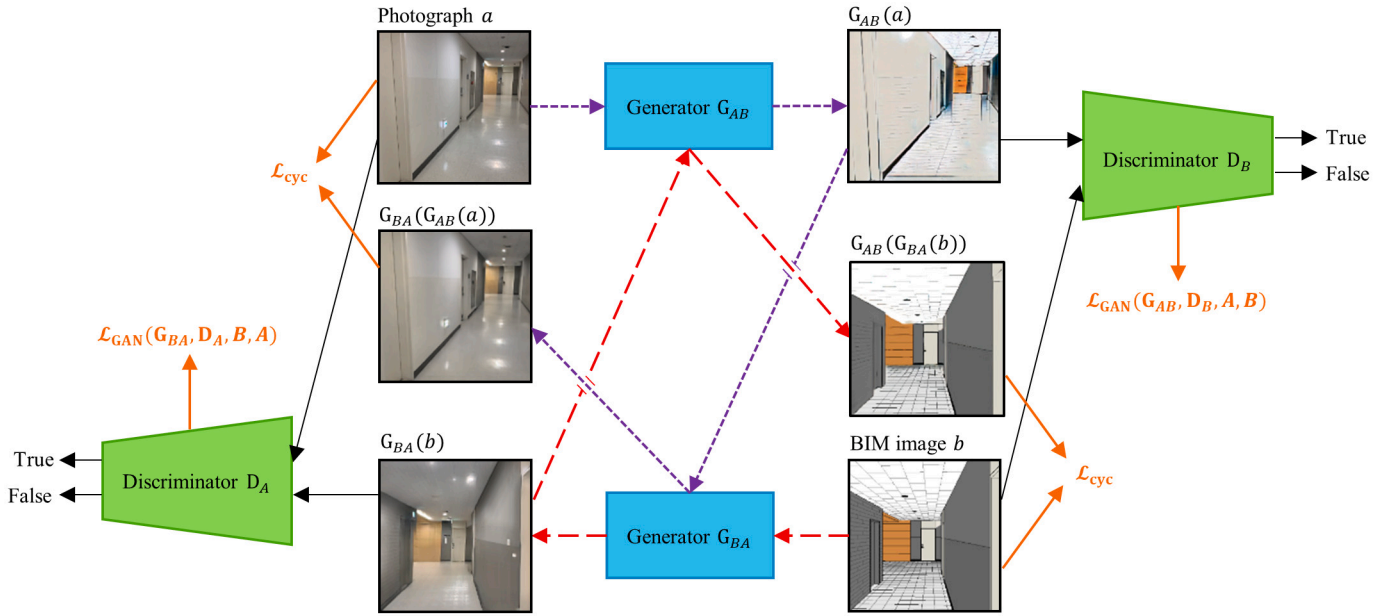


Fig. 2. Framework for training CycleGAN.

the GAN, image-to-image translation, which transforms an image into another image with a different representation, was performed. Isola et al. [18] proposed the Pix2pix model, in which an input condition was added to the loss function of the GAN to create an output image related to the input image. Whereas a basic GAN maps random noise vector z to output image y , the generator of Pix2pix creates y using z and input image x , and the discriminator determines whether y is both a real image and an image created from x . Pix2pix successfully solves the image-to-image translation problem in various domains, but it can only be trained when paired images exist. CycleGAN [19] is a model that enables translation between two domain images using unpaired training data. It was found to generate more natural-looking images in comparison with the algorithm presented in [32].

In the construction field, research using the GAN is being conducted to synthesize images to provide training data for deep learning models. Bang et al. [34] used a GAN as a tool to create augmented data for the detection of construction resources. In their study, inpainting was performed using a GAN to generate various object placement patterns by adding new objects to the construction images. Gao et al. [35] created a labeled structural image dataset using DCGAN and the leaf-bootstrapping method. The resulting dataset was used to train a model that classifies the scene levels and damage conditions of structural images. Baek et al. [36] improved the performance of construction equipment classifiers using a GAN and adversarial training. They designed a model that combines a generator, discriminator, and classifier so that the classifier can be trained using adversarial samples. The model improved the classifier performance by over 4% on average by addressing the issue of weak classification boundaries. Hong et al. [37] proposed the initial idea of using CycleGAN to generate photorealistic synthetic data from BIM images for semantic segmentation at an academic conference. This paper strengthens the original and add new contents of the conference paper for academic rigor.

Although previous studies have advanced the state of the art in synthetic data generation with regard to the construction industry, there

has been a lack of studies to address the difference between synthetic data and real-world data. A domain adaptation is a technique to teach a shift between source domain data and target domain data so that a model trained with source data performs well with target data. Among domain adaptation methods, the adversarial-based approach can be divided into generative models and non-generative models [38]. Generative models include Pix2pix, which create simulated data similar to target domain data based on GAN. Non-generative models adopt a method of simultaneous training for the main task and domain confusion so that the feature extractor derives the discriminative representation and the domain-invariant representation. Ganin et al. [39] proposed a Gradient Reversal Layer that is added to the feed-forward model to form a non-generative model for the effect of adaptation behavior. In the domain adaptation context, CycleGAN is categorized as the adversarial reconstruction approach that reconstructs source or target samples through cyclic mapping using GAN to improve domain adaptation performance.

We propose a methodology for generating synthetic data regarding infrastructure scenes based on both 3D models and actual images. The proposed methodology creates a mapping function that transforms BIM images into real-world images through CycleGAN training. The mapping function transforms BIM images such that they become photorealistic while maintaining their structure. Therefore, the synthetic data have the spatial features of real-world images and contain labels for all the components of the infrastructure in the images. These synthetic data have excellent potential to train deep learning models for analyzing infrastructure scene in real-world images.

3. Methodology

The procedure for generating synthetic data is divided into three steps. The first is to train CycleGAN to enable translation between photographs and BIM images. The second is to create labeled synthetic images that are similar to photographs from the BIM images using the

generators of the trained CycleGAN. The last is to splice synthetic images together in order to create effective synthetic data in training deep learning models for infrastructure scene understanding. The following paragraphs describe the details of these steps.

3.1. Image-to-image translation

CycleGAN is a network that learns the mapping function between the source domain A and the target domain B . The network consists of two generators and two discriminators. CycleGAN is trained through adversarial loss [16], $\mathcal{L}_{\text{GAN}}(G_{AB}, D_B, A, B)$, as expressed by Eq. (1):

$$\mathcal{L}_{\text{GAN}}(G_{AB}, D_B, A, B) = E_{b \sim p_{\text{data}}(b)} [\log D_B(b)] + E_{a \sim p_{\text{data}}(a)} [\log (1 - D_B(G_{AB}(a)))], \quad (1)$$

where a and b represent the training samples $\{a_i\}_{i=1}^N$ in domain A and $\{b_j\}_{j=1}^M$ in domain B , respectively, with subscripts omitted. The generator G_{AB} is a network that translates a into b , and generator G_{BA} translates b into a . The discriminator D_A distinguishes between a and a fake a created by the generator, and D_B plays the same role for b . Through adversarial loss, G_{AB} is trained to make $G_{AB}(a)$ similar to b , and D_B is trained to better distinguish between b and $G_{AB}(a)$. $\mathcal{L}_{\text{GAN}}(G_{BA}, D_A, B, A)$ operates similarly.

Cycle-consistency loss [40] was applied because a network with only adversarial losses could generate a specific image characteristic that deceives the discriminator while ignoring the input image. With regard to the expression for cycle-consistency loss $\mathcal{L}_{\text{cyc}}(G_{AB}, G_{BA})$, represented by Eq. (2), $G_{BA}(G_{AB}(a_i))$ and $G_{AB}(G_{BA}(b_j))$ are trained to have the same values as a_i and b_j , respectively. That is, G_{AB} and G_{BA} are derived to have an inverse functional relationship with each other:

$$\mathcal{L}_{\text{cyc}}(G_{AB}, G_{BA}) = E_{a \sim p_{\text{data}}(a)} [\|G_{BA}(G_{AB}(a)) - a\|_1] + E_{b \sim p_{\text{data}}(b)} [\|G_{AB}(G_{BA}(b)) - b\|_1] \quad (2)$$

The loss function of CycleGAN is summarized by Eq. (3), and the model is trained to produce G_{AB}^* and G_{BA}^* that satisfy Eq. (4):

$$\mathcal{L}(G_{AB}, G_{BA}, D_A, D_B) = \mathcal{L}_{\text{GAN}}(G_{AB}, D_B, A, B) + \mathcal{L}_{\text{GAN}}(G_{BA}, D_A, B, A) + \lambda \mathcal{L}_{\text{cyc}}(G_{AB}, G_{BA}) \quad (3)$$

$$G_{AB}^*, G_{BA}^* = \min_{G_{AB}, G_{BA}, D_A, D_B} \max \mathcal{L}(G_{AB}, G_{BA}, D_A, D_B) \quad (4)$$

In this study, the mapping between photographs and BIM images was performed using CycleGAN. The training framework of CycleGAN is depicted in Fig. 2. The generator G_{BA} receives the BIM image b as an input and generates $G_{BA}(b)$ similar to the photograph a as an output. The generator G_{AB} receives $G_{BA}(b)$ as an input and generates an image $G_{AB}(G_{BA}(b))$ similar to b , which is the original BIM image. The generation process of G_{BA} and G_{AB} is represented by the dotted red arrows. Similarly, $G_{AB}(a)$ and $G_{BA}(G_{AB}(a))$ are generated from G_{AB} and G_{BA} for the photograph a , as represented by the dotted purple arrows in Fig. 2. With regard to the cycle-consistency loss, the differences between b and $G_{AB}(G_{BA}(b))$ and a and $G_{BA}(G_{AB}(a))$ are calculated using the L1-norm. Discriminator D_A determines whether input photograph a and fake photograph $G_{BA}(b)$ are real; similarly, discriminator D_B determines the authenticity of b and $G_{AB}(G_{BA}(b))$. Using adversarial loss $\mathcal{L}_{\text{GAN}}(G_{BA}, D_A, B, A)$, D_A is trained to distinguish between real and fake photographs, and thereafter, G_{BA} is trained to create images that deceive D_A , that is, images that are similar to the real photographs, using the input BIM images, each of which is represented by b . The process of

calculating the loss amount is represented by orange arrows in Fig. 2. Further, the black arrows represent the inputs to the discriminators and their outputs. As a result, we obtain G_{AB} and G_{BA} that enable mapping between photographs and BIM images while maintaining the structure so that the resultant image can be reverted to the original image.

3.2. Synthetic data generation

Synthetic images are created using a generator G_{BA} trained to generate BIM images in a photographic style. As the training of CycleGAN progresses, the weights of G_{BA} change, which affects the creation of synthetic images. Therefore, to obtain the G_{BA} that generates synthetic images that are the most suitable for instance segmentation, G_{BA} was stored for every 20 epochs out of 200 epochs and used to create synthetic images. As depicted in Fig. 1(b), when the BIM image is fed as an input to G_{BA} , a synthetic image is created in a photographic style. Because the structure of the BIM images is maintained in the synthetic images, the annotation of the BIM images stored in the BIM software is transferred to the synthetic images as it is.

3.3. Splicing synthetic datasets

A new algorithm, splicing synthetic datasets, is designed to enable “two-stage synthetic image generation” to make the best use of GAN-based models in the context of infrastructure scene understanding. The algorithm is to address a critical problem that the quality of synthetic images cannot be guaranteed despite the training epoch of CycleGAN is increasing. In experiments, we observed that the segmentation performance fluctuates for each synthetic dataset generated by different training epochs of CycleGAN with respect to most of target object classes. It is hard to decide when to stop training CycleGAN to get the best synthetic images for all classes. It is conjectured that a lack of visual features of the target infrastructure classes resulted in such performance variations.

To address the problem, the new algorithm selects the best synthetic images by comparing the performance of Mask R-CNN in each training epoch of CycleGAN. A key idea is to produce the best synthetic image dataset by combining synthetic object masks from multiple synthetic datasets generated from different CycleGAN models and assigning them to the background synthetic images. The details are as follows: First, CycleGAN generates a number of synthetic datasets with different hyper-parameters of a training epoch, a learning rate, and a decay rate. Second, Mask R-CNN is separately trained using each synthetic dataset, and the best model is determined by AP for all classes. The synthetic dataset used for training the best model is selected. This dataset will be used as the background images. Third, the best model for each class is determined based on AP; for the synthetic dataset used in training the best model, pixels corresponding to the target object class are extracted. Fourth, the extracted pixels of a synthetic image are overlaid to a background image. Since each synthetic image has a corresponding BIM image, a final synthetic image has the same scene structure as its corresponding BIM image. But the difference is that the final image is composed of one background image and multiple object masks (extracted pixels) from different synthetic image datasets. By this composition, the final images have better visual representations for target infrastructure classes that are advantageous to train Mask R-CNN. Algorithm 1 elaborates the implementation details.

Algorithm 1: Splicing synthetic datasets

Input: Synthetic datasets: $\{S_i\}_{i=1}^n$ and there are total m number of synthetic images in S_i , Instance segmentation models trained with S_i : $\{h_i\}_{i=1}^n$, The number of object classes: k , Performance of i^{th} instance segmentation model for all classes: $performance_{total}(h_i)$, Performance of i^{th} instance segmentation model for j^{th} class: $performance_{class_j}(h_i)$

Output: A final synthetic dataset: S_{final}

```

1  $i^* \leftarrow \operatorname{argmax}_i performance_{total}(h_i)$ 
2  $S_{background} \leftarrow S_{i^*}$ 
3 for  $j \leftarrow 0$  to  $k$  do
4    $l \leftarrow \operatorname{argmax}_i performance_{class_j}(h_i)$ 
5   for  $p \leftarrow 0$  to  $m$  do
6     Overlay object masks of  $j^{th}$  class of  $S_l[p]$  on top of  $S_{background}[p]$ 
7   end for
8 end for

```

Table 1

Training dataset of Mask R-CNN for instance segmentation of infrastructure scenes.

Name	Type	Number
Photograph	Photographs	100
BIM	BIM images	100
Synthetic20	Synthetic data	100
Synthetic40		
Synthetic60		
Synthetic80		
Synthetic100		
Synthetic120		
Synthetic120d		
Synthetic140		
Synthetic140d		
Synthetic160		
Synthetic160d		
Synthetic180		
Synthetic180d		
Synthetic200		
Synthetic200d		
SUN database	Photographs	10,000

4. Experimental study

Synthetic data generation, and instance segmentation for synthetic data evaluation were performed using data acquired from the first engineering building of Yonsei University and Jungnang Railway Bridge located in Seoul, Korea. For the instance segmentation of the building scene, five classes were selected: door, wall, floor, ceiling, and window. Four classes were selected for the instance segmentation of the bridge scene: slab, girder, coping, and column. Synthetic datasets were evaluated by training the instance segmentation network (Mask R-CNN) using photographs, a BIM dataset, and synthetic datasets and testing the network using a photograph test dataset. Herein, “synthetic datasets” refer to datasets generated from G_{BA} stored every 20 epochs in CycleGAN and the dataset made by splicing them; each dataset was used to train different Mask R-CNNs.

4.1. Mask R-CNN

Mask R-CNN is a model developed from R-CNN [41], an object detection model. In R-CNN, object detection is performed through two steps: 1) A region of interest is extracted through selective search; 2)

extracted region proposals are passed through a CNN and the region determined as an object is regressed and classified. Subsequently, Fast R-CNN [42], which reduced the amount of computation required by adding a region of interest (RoI) pooling layer to R-CNN, was developed. Fast R-CNN has evolved into Faster R-CNN [43], which exhibits improved computational speed by replacing selective search with a region proposal network. Mask R-CNN is a model with a layer added to Faster R-CNN that determines whether each pixel inside a bounding box belongs to an object or not. In Mask R-CNN, errors occurring in RoI pooling have also been reduced by aligning the RoI. At the time of its publication, Mask R-CNN, with ResNeXt-101-FPN set as the backbone, achieved the highest instance segmentation performance among the existing single models. Currently, Mask R-CNN, with SpineNet [44] as the backbone, ranks third in instance segmentation performance on the COCO test-dev dataset [45]. Therefore, we concluded that the Mask R-CNN architecture is still promising for object recognition tasks and adopted it as a desirable model to verify the effectiveness of the generated synthetic data.

4.2. Implementation details

The proposed methodology was implemented using Python on the Ubuntu 16.04 operating system with a Geforce RTX 2080 Ti GPU and an Intel Core i5-7600 CPU. The deep learning libraries Keras [46] and TensorFlow [47] were used to implement both CycleGAN and Mask R-CNN. The batch size of CycleGAN was set to 1, and λ , which is the weight of the cycle-consistency loss expressed in Eq. (3), was set to 10. Two learning rates were applied to train CycleGAN for a total of 200 epochs: the first learning rate was set to a constant value of 0.0002 for 200 epochs (constant case), and the second learning rate was set to 0.0002 for the first 100 epochs and linearly decreased from 0.0002 to zero for the remaining 100 epochs (decay case).

The configuration of Mask R-CNN was set with the backbone of ResNet-101 [48] and a batch size of 2. We initialized the Mask R-CNNs with COCO pre-trained weights. We applied a learning rate of 0.001 on the first epoch and a learning rate of 0.0001 from the second epoch, and trained the models for 1000 epochs for all training datasets. When selecting synthetic datasets in Algorithm 1, there could be a case when the best performance of Mask R-CNN for a certain class is the same for different synthetic training datasets. In such case, a dataset with the higher value of AP for all classes is selected.

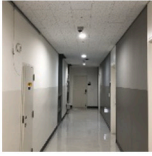
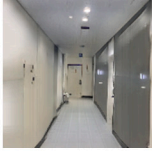
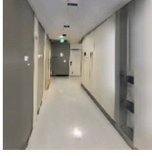
No.	Dataset	Images			Labels
1	Photograph (100)				.json
2	BIM (100)				color mask
	G_{BA} ↓				
3	Synthetic20 (100)				color mask

Fig. 3. Sample data from the Mask R-CNN training datasets.



Fig. 4. Annotation example of photographs.

4.3. Experiment on building elements

4.3.1. Datasets for training CycleGAN

To train CycleGAN, photographs and BIM images of the first engineering building of Yonsei University were used. The region of interest for data acquisition was an L-shaped corridor approximately 32 m long and 1.8 to 2.5 m wide. The BIM images were extracted from the BIM implemented using the Revit software [49]. For the photographic data required to train CycleGAN, 550 photographs were captured from the perspective of a pedestrian in the corridor. In this set of photographs, those of size 3024×3024 were reduced to a size of 512×512 to decrease the required computational power. For the BIM data required to train CycleGAN, 564 BIM images of size 512×512 were acquired by creating 3D views in Revit and extracting the views as images, leading to the creation of BIM images.

4.3.2. Datasets for mask R-CNN

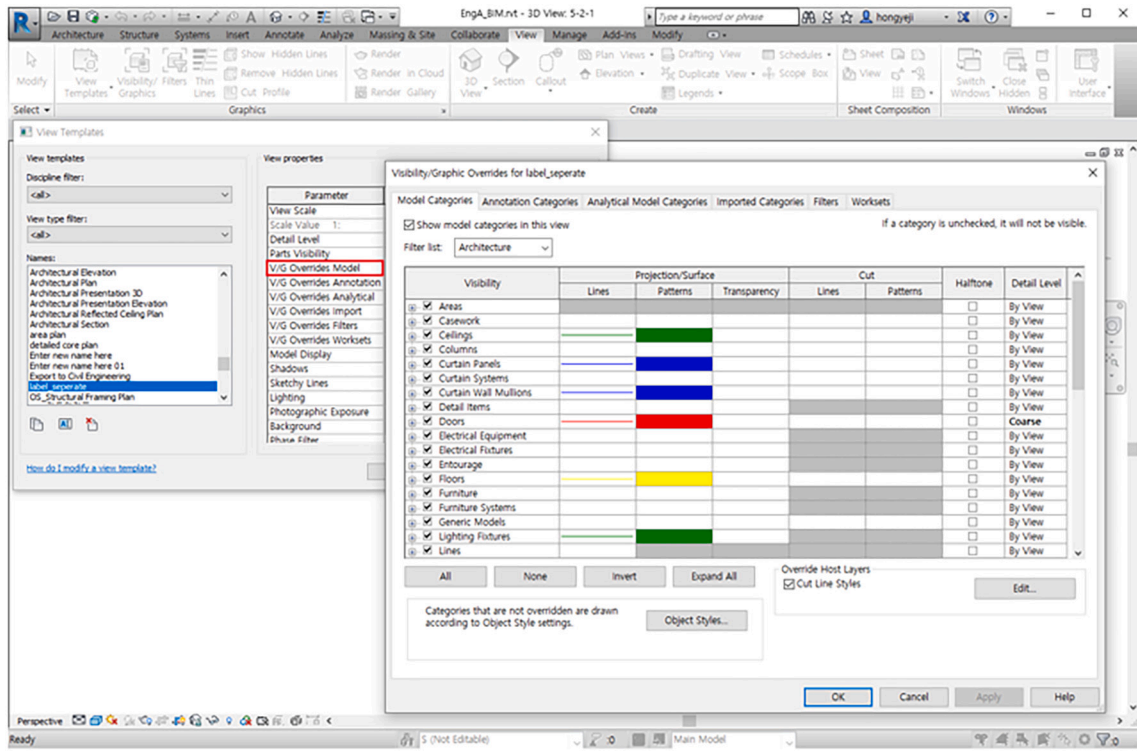
The Mask R-CNNs were trained on 18 datasets, which are presented in Table 1. Among them, 17 datasets consisted of 100 images, each of size 512×512 , and the other dataset consisted of 10,000 images of size 512×512 . The 17 datasets were as follows: a dataset of photographs (Photograph in Table 1), a dataset of BIM images (BIM in Table 1), and 15 datasets of synthetic images. The photographs, BIM images, and

synthetic images were all captured from the same position and angle as shown in Fig. 3.

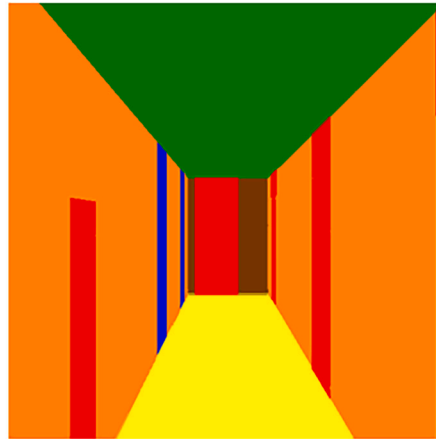
The synthetic datasets are denoted as “Synthetic” followed by the number of epochs corresponding to the generator of the CycleGAN from which the datasets were obtained. The suffix “d” is added to datasets created from the generator trained with the decaying learning rate. For example, the eighth and ninth datasets in Table 1, Synthetic120 and Synthetic120d, contain synthetic data generated based on the weights obtained in the 120th epoch using the generators trained with the constant and decaying learning rates, respectively. The datasets of synthetic images were obtained by feeding the BIM images to the generator G_{BA} obtained every 20 epochs during the training process of CycleGAN. Therefore, five generators were created during the first 100 epochs. Owing to the two different learning rates (constant case and decay case) mentioned in Section 4.2, 10 additional generators were created during the remaining 100 epochs, resulting in a total of 15 generators.

The 18th training dataset was acquired from the SUN database, a benchmark dataset consisting of 908 scene categories. The SUN database was selected as a training dataset to compare the building synthetic data with large-scale publicly available datasets. From the fully annotated 16,873 images of the SUN database, corrupt files and images that did not include the target classes were removed, and 10,000 images were randomly selected from the remaining images. The test dataset consisted of 100 photographs captured from the perspective of a pedestrian walking down the corridor while varying the position and angle of the camera.

The photographs were labeled manually using a VGG Image Annotator (VIA) [50]. As exhibited in Fig. 4. Annotation example of photographs, the labeling was performed by drawing a polygon for each instance and designating a class to it. The BIM images were labeled through the view template function of Revit, which can be employed to change the visibility and graphic settings of the view. As presented in Fig. 5(a), the visibility for each element category was set in the V/G Overrides Model parameter of the View Template menu so that the five classes could be classified by color. Views to which the view template was applied were extracted as 512×512 images (color masks) and used as annotations for the original BIM images. The synthetic images have the same content as the BIM images; therefore, the color masks of the BIM images were designated as the labels of the synthetic images. Finally, approximately 0.25% of the instances of the image area that



(a)

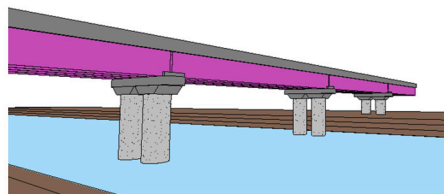


(b)

Fig. 5. Annotation example of BIM and synthetic images: (a) view template setting for creating color mask, (b) color mask example.



(a)



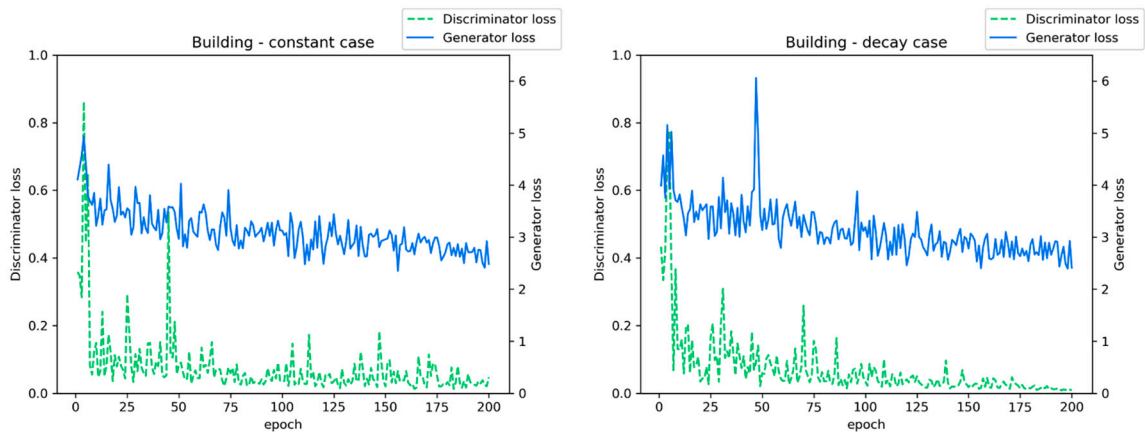
(b)

Fig. 6. Jungnang Railway Bridge: (a) the photograph of the whole view and (b) the corresponding BIM image.

Table 2

Performance of Mask R-CNNs trained on 18 datasets (%).

	AP _{door}	AP _{wall}	AP _{floor}	AP _{ceiling}	AP _{window}	AP _S	AP _M	AP _L	AP
Photograph	85.1	86.0	100	95.0	68.5	14.7	74.1	93.7	86.7
BIM	17.4	64.0	100	71.0	14.8	0.7	33.7	66.5	52.3
Synthetic20	37.5	72.8	100	86.0	11.1	2.6	47.7	75.5	63.7
Synthetic40	46.1	71.5	100	82.0	13.0	6.1	45.3	77.1	64.9
Synthetic60	75.1	67.5	100	84.0	22.2	6.6	54.7	80.9	71.0
Synthetic80	62.1	70.8	100	87.0	5.6	6.4	49.1	80.9	68.4
Synthetic100	66.5	74.8	100	84.5	11.1	8.4	54.4	80.5	71.7
Synthetic120	70.7	74.2	99.0	87.0	1.9	12.5	53.3	84.0	72.5
Synthetic120d	48.1	74.0	77.0	87.0	42.6	11.0	56.3	72.9	65.7
Synthetic140	47.6	74.5	97.0	87.0	13.0	8.2	50.4	77.7	67.4
Synthetic140d	53.1	76.9	98.5	88.5	3.7	6.6	51.8	81.0	69.7
Synthetic160	66.9	71.4	98.0	90.5	11.1	3.5	51.1	82.2	71.1
Synthetic160d	53.9	76.5	99.0	87.2	29.6	2.9	52.1	82.6	70.3
Synthetic180	58.4	72.0	99.0	91.0	7.4	8.8	51.8	79.7	68.9
Synthetic180d	51.1	76.9	98.0	87.0	1.9	15.4	50.7	79.5	69.0
Synthetic200	44.0	72.7	100	84.5	9.3	10.5	48.9	77.9	64.9
Synthetic200d	63.9	75.7	98.0	87.0	0	9.6	53.2	81.1	71.4
Avg(Synthetic)	56.3	73.5	97.6	86.7	12.2	7.9	51.4	79.6	68.7
SUN database	27.2	41.7	100	87.5	1.9	0	14.9	66.1	44.8

**Fig. 7.** The training loss curves for the building scenes (Left: loss curves for a constant learning rate; Right: loss curves for a changing learning rate).**Table 3**

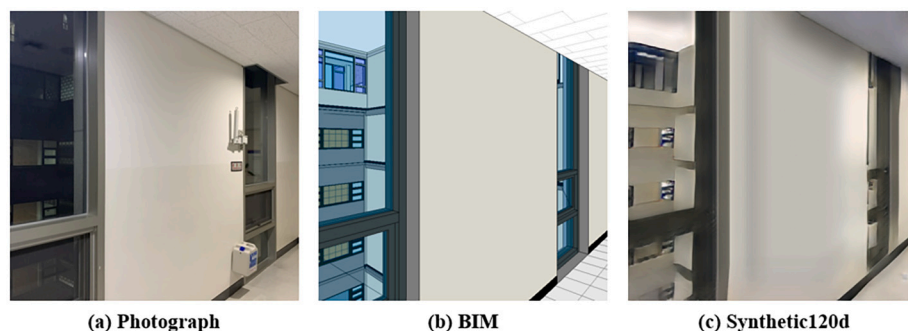
Number of instances by class and size in the building test dataset.

Door	Wall	Floor	Ceiling	Window	Small	Medium	Large
276	529	100	128	42	60	416	599

were smaller than 655 pixels were excluded from the label because even humans could not distinguish among them.

4.4. Experiment on structural elements of bridge

This experiment was conducted on the Jungnang Railway Bridge, a box girder bridge located in Seoul, Korea. Fig. 6 shows the whole view of the bridge and the corresponding BIM image. Among the photographs acquired within 50 m from the bridge, 300, 200, and 150 images were

**Fig. 8.** Comparison of windows in the different domain images.

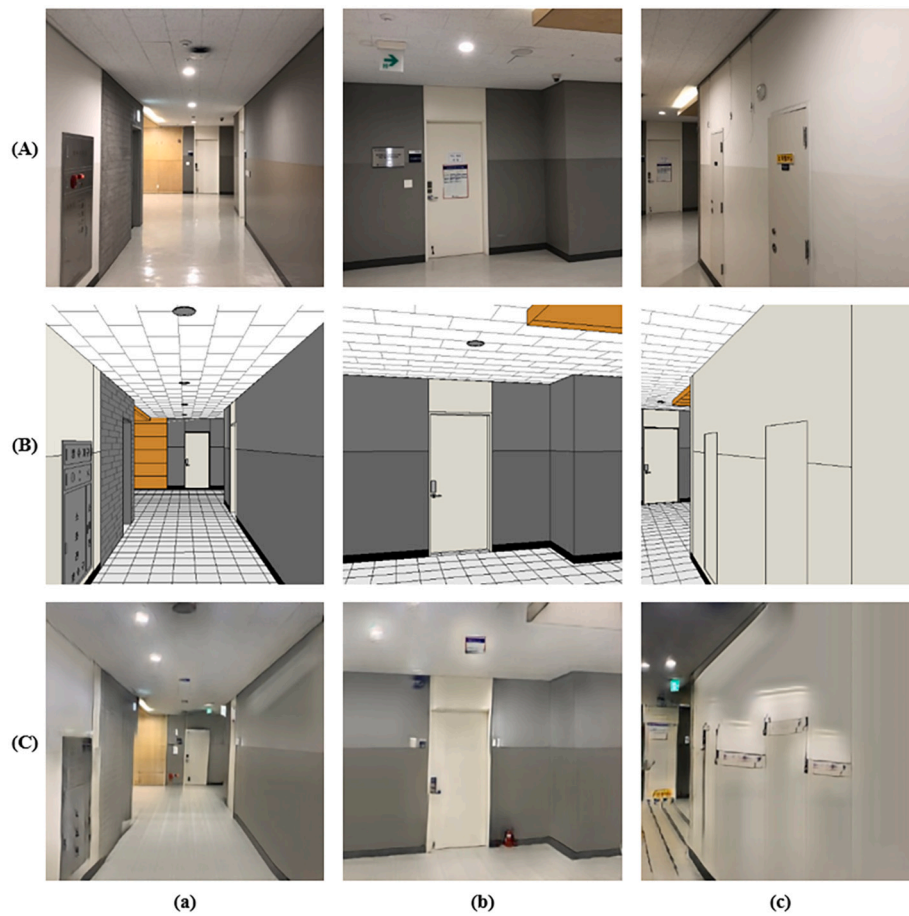


Fig. 9. Sample images of Photograph (A), BIM (B), and Synthetic120 (C).

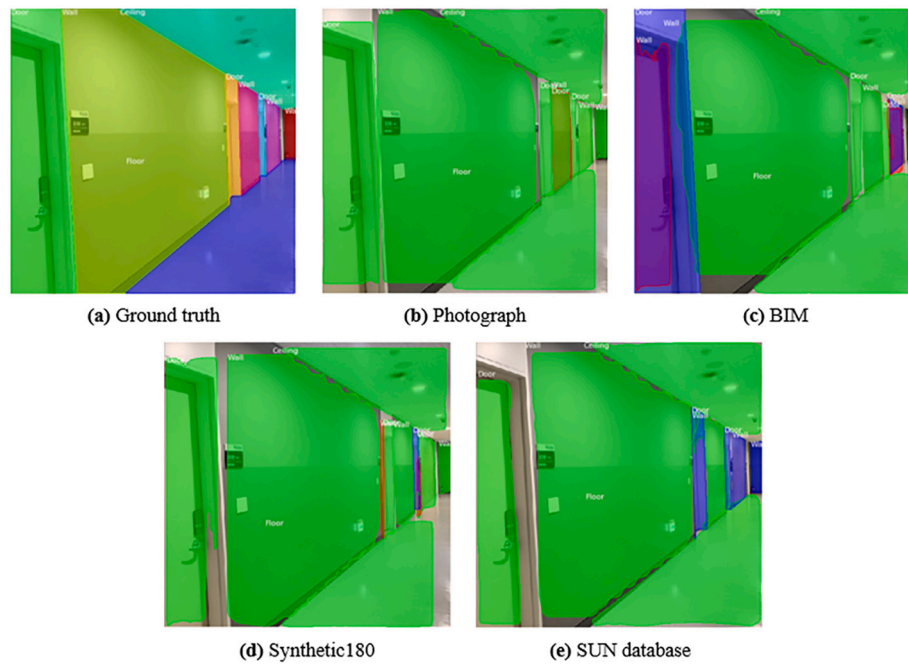


Fig. 10. Comparison of instance segmentation results obtained using different training datasets.

Table 4

Performance of Mask R-CNNs trained with the four additional datasets (%). The number in parentheses means the number of images randomly selected from the dataset.

	AP _{door}	AP _{wall}	AP _{floor}	AP _{ceiling}	AP _{window}	AP _S	AP _M	AP _L	AP
Photograph (5)	34.3	63.8	85.0	80.5	0	0	32.7	71.2	54.5
Photograph (5) & Synthetic120 (95)	64.8	78.7	100	89.0	31.5	8.3	54.0	85.8	74.5
Photograph (10)	60.2	73.7	97.0	74.0	18.5	6.6	54.0	78.9	68.2
Photograph (10) & Synthetic120 (90)	72.1	81.3	99.0	88.0	18.5	16.7	61.9	85.7	77.5

randomly selected and used as training data of CycleGAN, training data of Mask R-CNN, and test data of Mask R-CNN, respectively. 300 BIM images for training of CycleGAN and 200 BIM images for training of Mask R-CNN were acquired from the Revit walkthrough video, where the path and pose were similarly set as when taking the photographs. All photographs and BIM images used in the experiment were cropped and resized to 512×512 size. As in the experiment on building elements, 15 synthetic datasets were acquired using the generators of CycleGAN obtained every 20 epochs and the BIM images for the training of Mask R-CNN. As aforementioned, four main structural elements were used as the classes for scene analysis of the girder bridge: slab, girder, coping, and column. The slab is a plate-shaped structural member that forms the floor of a bridge, and the girder is a beam that supports the slab. The girder is in the form of a box made of steel in this bridge. The copings and columns are substructures of the bridge which support the load of the upper structure.

4.5. Quantitative measure: AP

Average precision (AP) was used as a quantitative measure to evaluate the performance of the Mask R-CNNs. For each image, the AP was calculated in the same way as in PASCAL Visual Object Classes Challenge 2010 (VOC2010) [51]. We evaluated the arithmetic mean of the APs of all the images to assess the instance segmentation performance. The value of the AP is between 0 and 1. The more accurate the predictions, the closer the AP is to 1.

5. Experimental results and discussion

5.1. Experimental results and discussion on building elements

Table 2 illustrates the performance of the Mask R-CNNs trained on the 18 training datasets for the building scenes. The training loss curves for the generator and discriminator are drawn in Fig. 7. The APs of the model trained with each training dataset are displayed for each class and size of the instance. The last column shows the APs for all instances. The Avg(Synthetic) row is the arithmetic mean of all the synthetic dataset rows. AP_S, AP_M, and AP_L are the APs corresponding to instances with small, medium, and large sizes, respectively. The criteria used to classify the sizes of the instances were set as 1024 (32^2) and 9216 (96^2), which follows COCO test-dev metrics [52]. Table 3 shows the number of instances by class and size of the test dataset. As datasets 1–17 represent the same scene, the differences in performance among them are attributed to the differences in the image styles and translation capabilities of the generators.

Table 2 indicates that all the models trained on the synthetic dataset achieved better instance segmentation performance than that of the model trained on the BIM. The model trained on the BIM had an AP of 52.3%, whereas the models trained on the synthetic datasets yielded an AP between 63.7% (Synthetic20) and 72.5% (Synthetic120). A difference of at least 11.4 percentage points and a maximum difference of 20.2 percentage points in performance between the results obtained using the BIM and synthetic data indicate that the synthetic data successfully simulated the features of the photographs.

In the Avg(Synthetic) column of Table 2, the AP for the window class is significantly lower than the Photograph column. As seen from the windows in Fig. 8, the exterior lighting condition of a photograph in

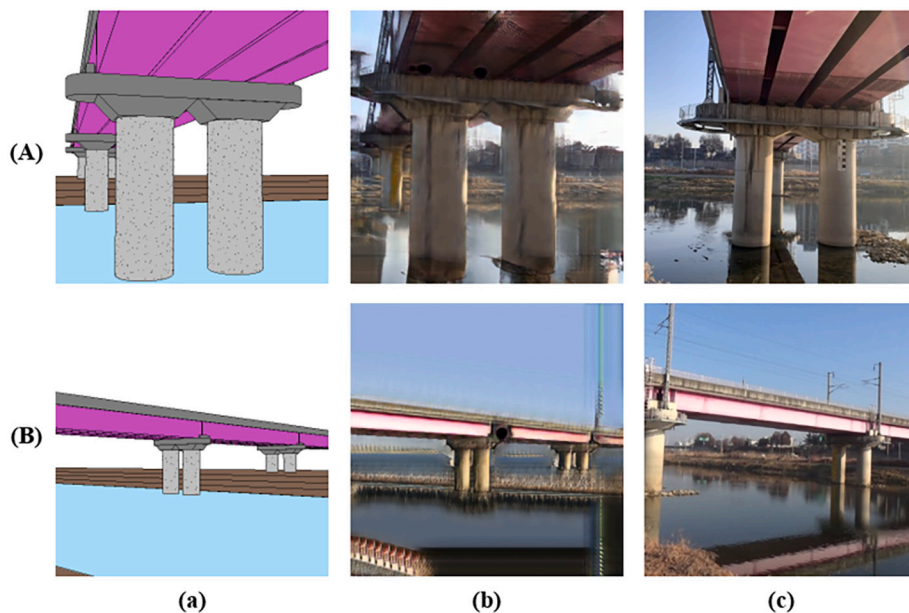


Fig. 11. Impact of the distance between the object and the camera on the quality of synthetic data; rows (A) and (B) show images from a relatively short distance and long distance, respectively; columns (a), (b), and (c) indicate the BIM, Synthetic200, and Photograph datasets, respectively.

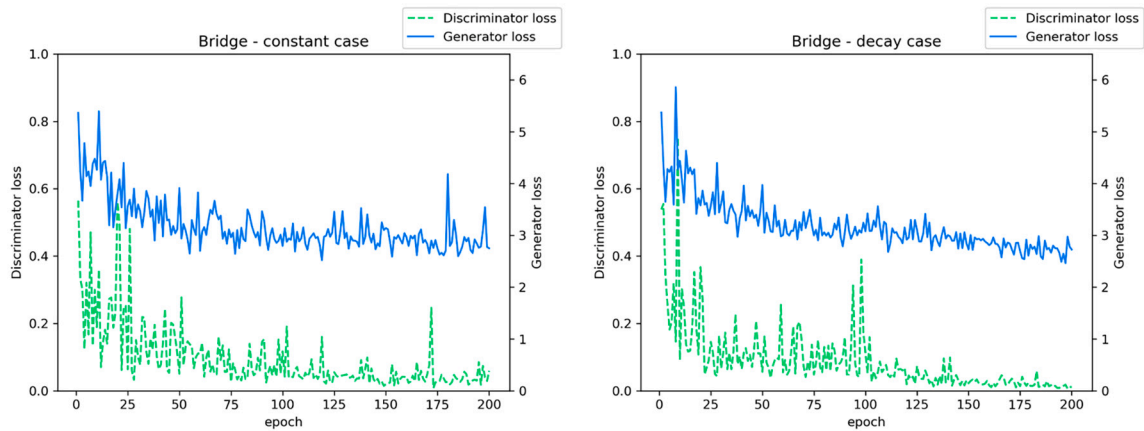


Fig. 12. The training loss curves for the bridge scenes (Left: loss curves for a constant learning rate; Right: loss curves for a changing learning rate).

Table 5

Performance of Mask R-CNNs trained with 17 datasets on the bridge scenes (%).

	AP _{slab}	AP _{girder}	AP _{coping}	AP _{column}	AP _S	AP _M	AP _L	AP
Photograph	85.5	96.8	97.8	98.2	71.5	89.0	94.5	95.0
BIM	21.2	39.3	11.6	30.8	0	13.5	31.5	23.0
Synthetic20	38.3	60.9	53.2	79.0	15.7	56.6	53.0	59.9
Synthetic40	48.5	81.7	62.7	88.0	24.7	62.5	75.8	71.6
Synthetic60	55.6	87.9	77.8	89.6	30.0	69.2	84.2	78.8
Synthetic80	46.8	91.9	77.3	86.4	28.4	68.5	81.2	77.8
Synthetic100	68.5	89.3	74.8	87.5	26.1	68.2	81.9	79.5
Synthetic120	54.0	91.8	80.4	91.9	30.3	72.0	84.5	81.9
Synthetic120d	41.5	90.0	76.1	92.8	25.2	70.2	78.7	78.5
Synthetic140	46.0	90.5	77.3	90.5	26.7	70.2	78.1	78.1
Synthetic140d	35.5	91.0	76.5	92.1	34.6	69.8	79.7	77.6
Synthetic160	57.3	89.6	78.3	92.8	34.8	71.5	84.7	81.4
Synthetic160d	35.1	87.9	78.5	87.1	20.6	66.6	77.9	75.2
Synthetic180	61.3	90.0	80.2	90.1	23.5	72.0	85.9	81.8
Synthetic180d	32.7	86.0	77.1	92.4	29.1	70.7	76.4	76.2
Synthetic200	47.2	87.7	87.3	94.5	32.6	74.2	81.2	82.3
Synthetic200d	40.3	88.3	78.6	92.7	32.3	72.0	76.5	78.7
Avg(Synthetic)	47.2	87.0	75.7	89.8	27.6	68.9	78.6	77.3

Fig. 8(a) is different from those of a BIM image in Fig. 8(b) and a synthetic image in Fig. 8(c). In other words, the generator G_{BA} failed to represent the darkness in the windows of the synthetic images. There are two reasons why generators failed to represent darkness. The first reason is that among the training data of CycleGAN, the number of images, including the window, was smaller than that of other classes. The second reason is that the photograph was taken at a night time whereas the BIM image is captured in a daytime lighting setting; as can be seen in Fig. 8, what is seen through the window of the BIM image is represented as a dark area in the photograph. As a result, we consider that Mask R-CNN was not trained enough with respect to the characteristics of the real window when it was trained on the synthetic datasets.

The synthetic dataset that yielded the best performance indicated a 14.2% lower performance than that derived from the Photograph. This difference in performance can be attributed to the remaining difference in the characteristics of the synthetic data compared to those of the original photographs, although this deviation has been significantly reduced. Fig. 9 presents sample images corresponding to the Photograph, BIM, and Synthetic120. From Fig. 9(a), it is confirmed that a part of the image texture is not well reflected in the synthetic image. The grid pattern on the floor, which is hardly noticeable in the actual photograph, is clearly visible in the synthetic image. Fig. 9(b) indicates that small objects that are present in the photograph, but not in the BIM image, appeared at inappropriate locations in the synthetic image. For example, the nameplate that should be attached to the door is located on the ceiling in the synthetic image. In Fig. 9(c), the aforementioned object

Table 6

Number of instances by class and size in the bridge test dataset.

Slab	Girder	Coping	Column	Small	Medium	Large
129	307	273	452	54	577	530

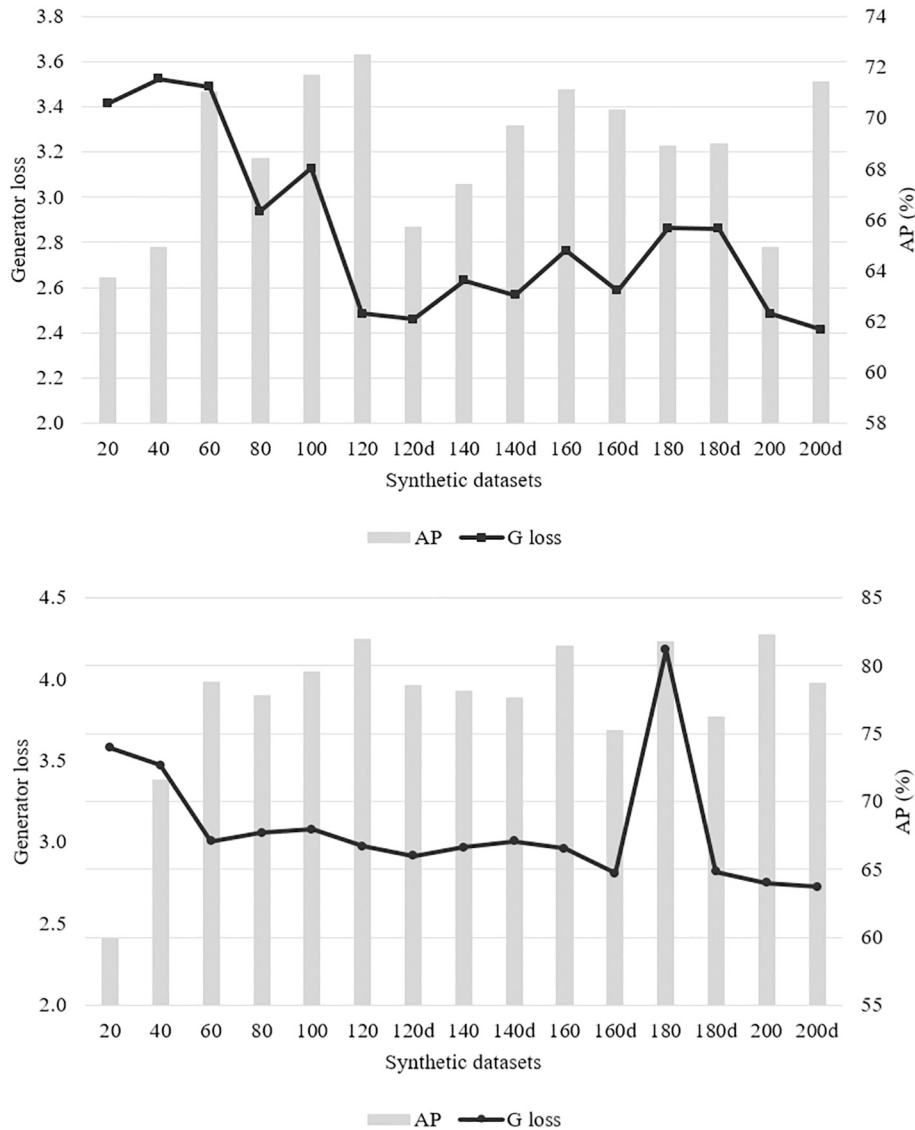
patterns appear repeatedly along the boundary line, making the boundaries between instances unclear. The architecture of the CycleGAN discriminator could be the reason for such frequent generation of objects that do not exist in the BIM images but exist only in photographs. For the CycleGAN discriminator network, 70×70 PatchGAN [18] was used; it is a network that determines whether an image is real or fake using the average of the probability that each local image patch is real. The strong locality seems to have contributed to the generation of improper small objects in the synthetic images.

The performance of the model trained on the SUN database was 44.8%, which was lower than that of the model trained on the BIM, 52.3%. When comparing the performance derived from the synthetic datasets and the SUN database in Table 2, the average AP of synthetic datasets in the door, wall, and window classes are 10 to 30 points higher than that of the SUN database. Although the SUN database contains 100 times more images and a wide variety of scenes, the synthetic datasets seem to better capture the characteristics of the three classes used in the experiment. The model trained with various features in the SUN database was not effective in precisely detecting instances of a specific appearance.

Table 7

Performance of Mask R-CNNs trained with the six bridge datasets (%). The number in parentheses means the number of images randomly selected from the dataset.

	AP _{slab}	AP _{girder}	AP _{coping}	AP _{column}	AP _s	AP _M	AP _L	AP
Photograph (5)	38.7	73.6	72.2	81.4	11.9	53.8	77.5	69.5
Photograph (5) & Synthetic200 (195)	66.1	91.1	90.3	95.3	37.5	77.9	89.4	87.3
Photograph (10)	71.0	82.1	82.8	93.6	21.2	71.1	90.2	83.3
Photograph (10) & Synthetic200 (190)	76.6	93.5	91.1	96.1	31.9	79.7	92.5	89.7
Photograph (20)	70.2	89.9	90.6	93.8	29.9	80.1	89.1	87.1
Photograph (20) & Synthetic200 (180)	77.0	94.2	93.0	96.2	46.4	81.3	91.4	90.7

**Fig. 13.** The generator loss and the AP for each synthetic dataset (Top: the results for the building scenes; Bottom: the results for the bridge scenes).

An example of the result of instance segmentation of building elements is shown in Fig. 10. In Fig. 10(a), the labels of all instances are displayed as different colors. In Fig. 10(b)–(e), the areas corresponding to correct and wrong predictions are displayed in green and red, respectively. The blue areas in Fig. 10(b)–(e) represent the regions that were not successfully identified by the Mask R-CNNs. Therefore, some areas exhibit a purple color because the areas of misclassification (represented by red) can overlap with the areas that were not identified (represented by blue).

In the images of the corridor, which were employed for testing the models in this experiment, instances with small areas, such as doors,

walls, and windows located far from the camera, often appear. In general, the models trained on the synthetic datasets did not detect such small-sized instances well as shown in Fig. 10(d). However, Synthetic180d achieved an exceptionally higher AP_s than one derived from the Photograph. This result suggests that synthetic data has the potential as effective training data that surpasses photographs.

We created four additional training datasets by combining a large amount of synthetic data and a small number of photographs. The first and third training datasets consisted of five and 10 images randomly selected from the Photograph. The second and fourth training datasets were constructed by adding 95 and 90 synthetic images from

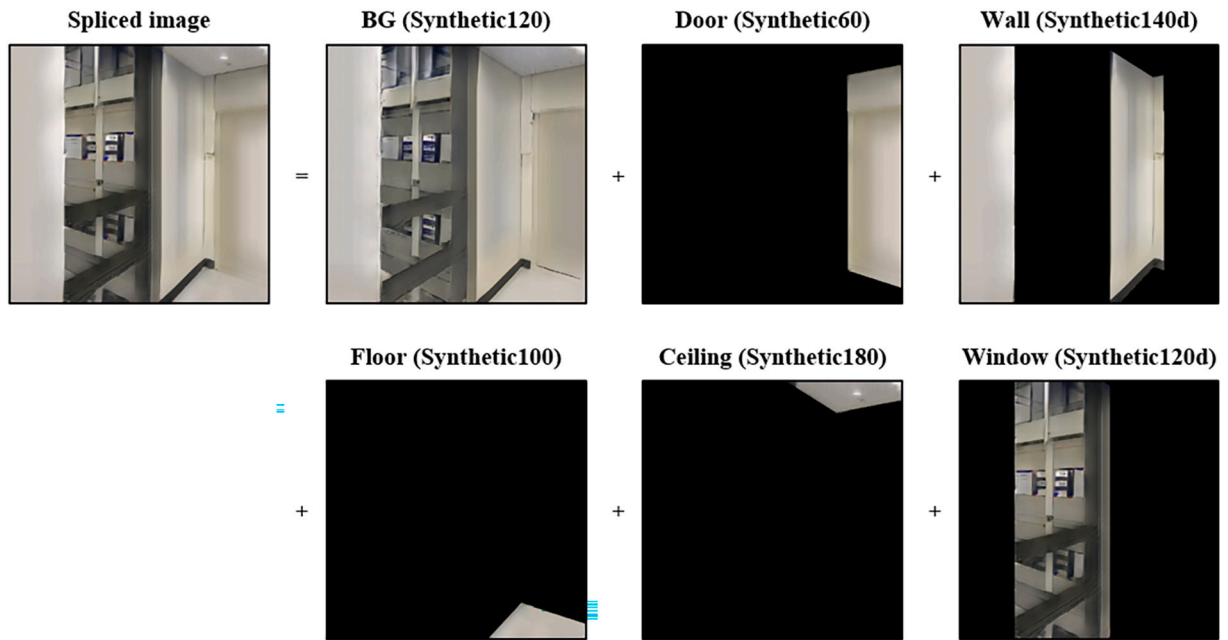


Fig. 14. Example of a spliced image and the components of it.

Table 8

Performance of Mask R-CNNs trained with the Spliced Synthetic dataset on the building scenes (%).

	AP _{door}	AP _{wall}	AP _{floor}	AP _{ceiling}	AP _{window}	AP _S	AP _M	AP _L	AP
Spliced Synthetic dataset	51.0	80.0	100	89.5	24.1	6.58	53.9	81.5	71.6

Table 9

Performance of Mask R-CNNs trained with the Spliced Synthetic dataset on the bridge scenes (%).

	AP _{slab}	AP _{girder}	AP _{coping}	AP _{column}	AP _S	AP _M	AP _L	AP
Spliced Synthetic dataset	63.3	93.2	83.2	93.5	36.9	75.5	87.5	84.9

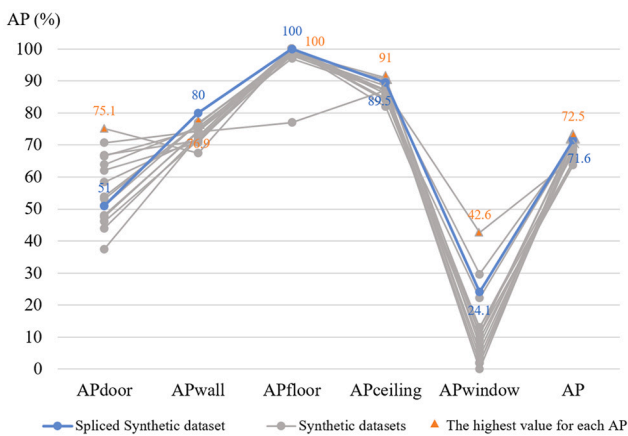


Fig. 15. Comparison of Mask R-CNNs performance derived from the Spliced Synthetic dataset and the other Synthetic datasets.

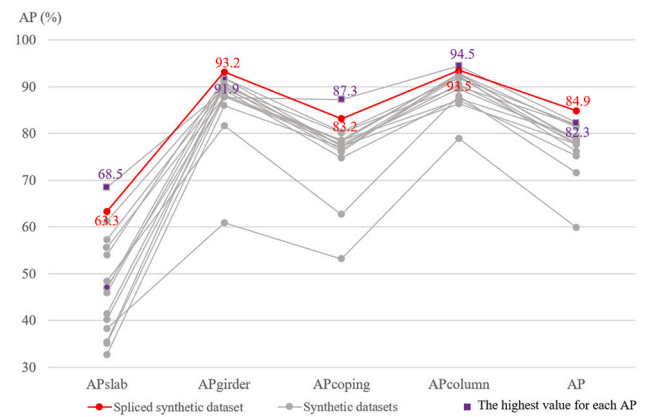


Fig. 16. AP graph of Mask R-CNNs trained with Spliced Synthetic dataset and the other Synthetic datasets on bridge scenes.

Synthetic120 to the first and third training datasets, respectively. The 95 and 90 synthetic images were selected such that they did not overlap with the five and 10 images from the Photograph, respectively. Table 4 shows the performance of the Mask R-CNN trained on the four datasets. The first training dataset did not contain any instance of the window class, so the AP for the window class was 0. The first training dataset did not sufficiently cover the characteristics of the space to be tested. However, with the addition of synthetic data, the information shortage was resolved and achieved the AP improvement of 20 percentage points. The fourth training dataset resulted in the AP of 77.5%. Although there was no improvement for the window class, the AP was 9.3 percentage points higher than the AP of the model trained on the third dataset. Even if only a small amount of photographs is used, it can be seen that when a large amount of synthetic data is added, the performance can be greatly improved. This performance improvement suggests that our methodology can reduce the time and cost required for data augmentation.

5.2. Experimental results and discussion on structural elements

Fig. 13 shows the impact of the distance between the object and the camera on the quality of Synthetic200 data. The synthetic image of Fig. 11(A)(b) created from the BIM image of Fig. 13(A)(a) is similar in texture, color, and pattern to the photograph of Fig. 11(A)(c). An example showing the high similarity to Fig. 11(A)(c) is the railing of the coping created in Fig. 11(A)(b). On the other hand, in the synthetic image of Fig. 11(B)(b) created from the BIM image of Fig. 11(B)(a), a relatively low level of similarity was observed, due to the distance between the camera and the subject. When the subject is captured as an image from a short distance, the subject occupies a large area in the image, and it is clear what the generator will do to create a realistic image. However, in the case of a long distance, the area occupied by the subject is small, so the generator is rather trained with the characteristics of the background and tries to create it. Therefore, in order to obtain good quality synthetic images, it is important to increase the area occupied by the subject in the image.

For the bridge scenes, the training loss curves for the generator and discriminator are drawn in Fig. 12. Table 5 shows the performance of Mask R-CNN trained with the Photograph, BIM, and Synthetic datasets. Table 6 shows the number of instances for each class and size included in the test dataset. Synthetic200 scored a maximum AP of 82.3%. This AP is 59.3 percentage points higher than the AP derived from the BIM (23.0%). In the Avg(Synthetic) row of Table 5, the AP for small-sized instances is notably lower than for the Photograph column. The poor segmentation performance for small-size instances could be attributed to the cycle-consistency loss, a constraint that allows images to retain their content after translation. The cycle-consistency loss is the L1-norm of the original image and the image that has passed through the mapping functions in both directions, that is, the sum of the differences between the pixel values of the two images. Therefore, because an instance with a small area exerts a low impact on the cycle-consistency loss, such an instance is more likely to lose its content when compared with an instance with a large area.

We created six training datasets for the structural elements of the bridge in the same way as the building experiment. The first, third, and fifth training datasets were created by randomly extracting five, 10, and 20 images from the Photograph. The second, fourth, and sixth training data were created by randomly selecting 195, 190, and 180 images from Synthetic200 and adding them to the first, third, and fifth dataset, respectively. Table 7 shows the performance of Mask R-CNNs trained with the six new datasets. Comparisons of the first and second, third and fourth, and fifth and sixth rows of Table 7 showed that the performance of the model trained on the dataset to which synthetic data was added was improved compared to the model trained only on the photographs. In particular, the sixth training dataset, which includes only 10% of the images in the Photograph, showed a more than 90% AP. This is only 4.3 percentage points lower than the AP of the model trained with the

Photograph. In other words, the instance segmentation model that performs well on the real-world domain data was implemented by using small amount of photographs and large amount synthetic data.

5.3. Discussion on the correlation between generator loss curves and AP

In Fig. 13, we displayed the loss curve of the generator with synthetic datasets generated from different training epochs of CycleGAN; the AP of Mask R-CNN trained with the synthetic datasets was also plotted to figure out the correlation between those two values. The upper graph and the lower graph are regarding the building and the bridge scenes, respectively. It is confirmed that the synthetic data generated from lower loss values do not guarantee the high performance of the instance segmentation model. This means that a well-trained generator does not always create synthetic data suitable for training instance segmentation models.

5.4. Experimental results and discussion on two-stage synthetic image generation

Fig. 14 illustrates a process of creating synthetic training data through image splicing. In this example, Synthetic120, which recorded the highest AP, was used as a background. The pixels corresponding to door, wall, floor, ceiling, and window from Synthetic60, Synthetic140d, Synthetic100, Synthetic180, and Synthetic120d were overlaid on top of the background to form a spliced image.

The Spliced Synthetic datasets were made by Algorithm 1 for the building and bridge scenes, respectively. The datasets were used to train Mask R-CNN and the test results are shown in Table 8 and Table 9. Figs. 15 and 16 showed the performance curves for target object classes.

Using the proposed splicing algorithm, Mask R-CNN model recorded AP of 71.6% for the building scenes and 84.9% for the bridge scenes, as shown in Figs. 15 and 16. The advantage of the two-stage synthetic image generation algorithm is clear in that Mask R-CNN shows the better performances on most target classes compared to the model achieved the best performance in the original experimental setting. For the building scenes, the segmentation performance with respect to four classes—wall, floor, ceiling, and window—was improved by the margin of 5.8%, 1.0%, 2.5%, and 22.2%, respectively. For the bridge scenes, the segmentation performance with respect to three classes—slab and girder—was increased by the margin of 16.1% and 5.5%, respectively. Moreover, the proposed algorithm achieved the best performance for the bridge scenes in terms of AP for all classes, recording 84.9%. Although the performance was not the highest in the building scenes, the proposed algorithm helped to secure the higher performance of Mask R-CNN on most of target classes (four out of five classes) compared to the previous best model. The findings demonstrate how GAN-based models can be used to improve the instance segmentation performance and be applied to situations when the number of training data is insufficient.

6. Conclusions

Contextual awareness of the civil infrastructure scene is a technology that enables various automated tasks in the construction industry using visual data. For contextual awareness using deep learning, the preparation of large labeled training datasets is essential. The non-availability of infrastructure scene images and the difficulties faced while annotating them are factors that hinder the development of intelligent models for infrastructure status understanding. To address this challenge, we developed a CycleGAN-based methodology to automatically generate labeled infrastructure images. The proposed method can generate synthetic image data based on image-to-image translation and BIM images to considerably improve the performance of instance segmentation for infrastructure monitoring.

The synthetic data generated using the proposed method for the infrastructure scene understanding were validated through experiments

on training Mask R-CNNs for building elements and the structural elements of a bridge. Through training the instance segmentation networks on synthetic datasets of the building elements, average precisions between 63.7% and 72.5% were derived, corresponding to performances that were significantly higher than those of the model trained on the BIM dataset (52.3%) and the SUN database (44.8%). In the experiment on structural elements of the bridge, the model trained on the synthetic dataset achieved an AP of 82.3%, an improvement of 59.3 percentage points over that of the model trained on the BIM dataset. By employing Algorithm 1, the instance segmentation performance was improved such that most target object classes have the better performance of class AP compared with the best performance of the previous model without using Spliced Synthetic datasets.

The contribution of this study to the body of knowledge is three-fold. First, a novel methodology was created to generate 2D labeled synthetic data, which can be effectively used for object detection of infrastructure scenes. The methodology drastically reduces the time and cost required for data augmentation of infrastructure images. Through our methodology, labeled synthetic images are efficiently obtained without image annotation effort by using the BIM images. It has been demonstrated in experimental studies that the instance segmentation performance significantly improves when the model is trained on a dataset including the synthetic data. Second, it was found that the domain difference between photographs and BIM images can be overcome through the style-transfer approach. To the best of our knowledge, there is no precedent for synthetic data generation on infrastructure scenes by translating 2D images extracted from BIM into photorealistic images. The generated synthetic data derived from the generators are powerful, with realistic features and labels; thus, they can be effectively used to train deep learning models for object detection in images of infrastructure such as buildings and bridges. Third, this study presents a novel method to select best synthetic images and how to splice them to create effective training data for instance segmentation models that can secure the high performance.

The experiment performed in this study demonstrated promising results; however, there are still some issues to be resolved. First, the image-to-image translation should be improved to reduce the differences between photographs and BIM images. This issue is expected to be addressed through the introduction of advanced GAN models. Second, a fully automated method should be developed for the annotation of infrastructure images. Complete automation of the labeling task would increase the economic value of the proposed method. With these upgrades, the proposed method is expected to aid in the acceleration of the digitalization of infrastructure monitoring.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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